

```
#python list
```

```
n_list=[2.9,3000,"hello","welcome to python list",True]
```

```
print(n_list)
```

```
n_list=["happy",[2,0,1,5]]
```

```
print(n_list[0][3])
```

```
print(n_list[1][2])
```

```
my_list = ["mouse", [8, 4, 6], ['a','b','c']]
```

```
print(my_list)
```

```
print(my_list[0])
```

```
print(my_list[1][1])
```

```
print(my_list[2][2])
```

```
#negative indexing in lists
```

```
my_list = ['p','r','o','b','e']
```

```
print(my_list[-1])
```

```
print(my_list[-5])
```

```
odd = [1, 3, 5]
```

```
odd.append(7)
```

```
print(odd)
```

```
odd.extend([9, 11, 13])
```

```
print(odd)
```

```
odd = [1, 3, 5]
```

```
print(odd + [9, 7, 5])
```

```
print(["re"] * 3)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
print(len(thislist))
```

```
#Remove Specified Item:
```

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.remove("banana")
```

```
print(thislist)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
for x in thislist:
```

```
    print(x)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
i = 0
```

```
while i < len(thislist):
```

```
    print(thislist[i])
```

```
    i = i + 1
```

```
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]  
  
thislist.sort()  
  
print(thislist)
```

```
thislist = [100, 50, 65, 82, 23]  
  
thislist.sort()  
  
print(thislist)
```

```
thislist=["orange","mango","kiwi","pineapple","banana"]  
  
thislist.sort(reverse=True)  
  
print(thislist)
```

```
thislist = [100, 50, 65, 82, 23]  
  
thislist.sort(reverse=True)  
  
print(thislist)
```

```
thislist = ["apple", "banana", "cherry"]  
  
mylist = thislist.copy()  
  
print(mylist)
```

```
thislist = ["apple", "banana", "cherry"]  
  
mylist=list(thislist)  
  
print(mylist)
```

```
fruits = ['apple','banana','cherry']
```

```
fruits.insert(1, "orange")
```

```
print(fruits)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.pop(1)
```

```
print(thislist)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.pop()
```

```
print(thislist)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
del thislist[0]
```

```
print(thislist)
```

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.clear()
```

```
print(thislist)
```

```
fruits = ["apple", "banana", "cherry", "apple"]
```

```
x = fruits.count("apple")
```

```
print(x)
```

```
fruits = [1, 4, 2, 9, 7, 8, 9, 3, 1]
```

```
x = fruits.count(9)
```

```
print(x)
```

```
lst = [10, 20, 30, 40]
```

```
# Create a copy of the list
```

```
new_lst = lst.copy()
```

```
print("Original List :", lst)
```

```
print("Copied List  :", new_lst)
```

```
lst = [10, 20, 30, 40, 50]
```

```
value = 30
```

```
position = lst.index(value)
```

```
print("List  :", lst)
```

```
print("Value  :", value)
```

```
print("Index  :", position)
```

```
lst=[10,20,30,40,50]
```

```
value=30
```

```
position=lst.index(value)
print("List :",lst)
print("Value:",value)
print("index:",position)
```

```
t=['a','b','c']
x=t.pop(1)
print(t)
```

```
print(x)
```

```
t=['a','b','c']
del t[1]
print(t)
```

```
t=['a','b','c']
t.remove('b')
print(t)
```

```
nums = [3, 41, 12, 9, 74, 15]
```

```
print (len(nums))
print (max(nums))
```

```
print (min(nums))
```

```
print (sum(nums))
```

```
print (sum(nums)/len(nums))
```

```
#tuple
```

```
my_data = ("hi","hello","bye")
```

```
print(my_data)
```

```
my_data2=(1,2.8,"hello world")
```

```
print(my_data2)
```

```
my_data4=((2,3,4),(1,2,"hi"))
```

```
print(my_data4)
```

```
my_data=("hi","hello","bye")
```

```
print(my_data)
```

```
print(my_data[0])
```

```
print(my_data[2])
```

```
my_data=(1,2,"kelvin",8.9)
```

```
print(my_data[-1])
```

```
print(my_data[-3])
```

```
my_data=(1,"steve",(11,22,33))
```

```
print(my_data[1][3])
```

```
print(my_data[2][1])
```

```
my_data=(1,[9,8,7],"world")
```

```
print(my_data)
```

```
my_data[1][2]=99
```

```
print(my_data)
```

```
my_data = (11,22,33,44,55,66,77,88,99)
```

```
print(my_data)
```

```
print(my_data[2:5])
```

```
print(my_data[:4])
```

```
print(my_data[4:])
```

```
print(my_data[4:-1])
```

```
print(my_data[:])
```

```
my_data=(11,22,33,44,55,66,77,88,99)
```

```
print(my_data)
```

```
print(22 in my_data)
```

```
print(2 in my_data)
```

```
print(88 not in my_data)
```

```
print(101 not in my_data)
```

```
my_tuple = ("apple","orange","grapes","banana")
```

```
for fruit in my_tuple:
```

```
    print(fruit)
```

```
tuple1,tuple2 = ('xyz','zara','abc'),(456,700,200)
```

```
print("max value element :",max(tuple1))
```

```
print("max value element :",max(tuple2))
```

```
my_dict={'name':'jack','age':26}
```

```
print(my_dict['name'])
```

```
print(my_dict.get('age'))
```

```
#dictionary
```

```
my_dict={}
```

```
print("1. Empty Dictionary:", my_dict)
```

```
my_dict = {1: 'apple', 2: 'ball'}
```

```
print("2. Integer Keys:", my_dict)
```

```
my_dict = {'name': 'John', 1: [2, 4, 3]}
```

```
print("3. Mixed Keys:", my_dict)
```

```
my_dict = dict({'apple', 2:'ball'})
```

```
print("4. Using dict():", my_dict)
```

```
my_dict = dict([(1, 'apple'), (2, 'ball')])
```

```
print("5. From Sequence:", my_dict)
```

```
pairs = [(1, 'apple'), (2, 'ball')]
```

```
my_dict = dict(pairs)
```

```
print(my_dict)
```

```
my_dict={'name':'jack','age':26}
```

```
my_dict['age']=27
```

```
print(my_dict)
```

```
my_dict['address']='downtown'
```

```
print(my_dict)
```

```
tinydict={'name':'john','code':6734,'dept':'sales'}
```

```
print(tinydict)
```

```
print(tinydict.keys())
```

```
print(tinydict.values())
```

```
squares={1:1,2:4,3:9,4:16,5:25}
```

```
print(squares.pop(4))
```

```
print(squares)
```

```
squares.clear
```

```
print(squares)
```

```
squares={1:1,3:9,5:25,7:49,9:81}
```

```
print(1 in squares)
```

```
print(2 not in squares)
```

```
print(49 in squares)
```

```
my_set = {1, 2, 3}
```

```
print(my_set)
```

```
my_set = {1.0, "Hello", (1, 2, 3)}
```

```
print(my_set)
```

```
my_set = {1, 2, 3}
```

```
my_set.add(4)
```

```
print(my_set)
```

```
my_set = {1, 2, 3}
```

```
my_set.remove(2)
```

```
print(my_set)
```

```
numbers = [1, 2, 2, 3, 4]
```

```
my_set = set(numbers)
```

```
print(my_set)
```

```
nums1 = {1, 2, 2, 3, 4, 5}
```

```
nums2 = {4, 5, 6, 7, 7, 8}
```

```
distinct_nums = nums1.union(nums2)
```

```
print("The union of two sets is: ", distinct_nums)
```

```
nums1 = {1, 2, 2, 3, 4, 5}
```

```
nums2 = {4, 5, 6, 7, 7, 8}
```

```
common_nums = nums1.intersection(nums2)
```

```
print("The intersection of two sets is:", common_nums)
```

```
nums1 = {1, 2, 2, 3, 4, 5}
```

```
nums2 = {4, 5, 6, 7, 8, 8}
```

```
nums3 = nums1.difference(nums2)
```

```
nums4 = nums2.difference(nums1)
```

```
print("nums1 - nums2: ", nums3)
```

```
print("nums2 - nums1: ", nums4)
```

```
A = {1, 2, 3, 4}
```

```
B = {3, 4, 5, 6}
```

```
print(A | B)
```

```
print(A & B)
```

```
print(A - B)
```

output

[2.9, 3000, 'hello', 'welcome to python list', True]

p

1

['mouse', [8, 4, 6], ['a', 'b', 'c']]

mouse

4

c

e

p

[1, 3, 5, 7]

[1, 3, 5, 7, 9, 11, 13]

[1, 3, 5, 9, 7, 5]

['re', 're', 're']

3

['apple', 'cherry']

apple

banana

cherry

apple

banana

cherry

['banana', 'kiwi', 'mango', 'orange', 'pineapple']

[23, 50, 65, 82, 100]

['pineapple', 'orange', 'mango', 'kiwi', 'banana']

[100, 82, 65, 50, 23]

['apple', 'banana', 'cherry']

['apple', 'banana', 'cherry']

['apple', 'orange', 'banana', 'cherry']

['apple', 'cherry']

['apple', 'banana']

['banana', 'cherry']

[]

2

2

Original List : [10, 20, 30, 40]

Copied List : [10, 20, 30, 40]

List : [10, 20, 30, 40, 50]

Value : 30

Index : 2

List : [10, 20, 30, 40, 50]

Value: 30

index: 2

['a', 'c']

b

['a', 'c']

['a', 'c']

6

74

3

154

25.666666666666668

('hi', 'hello', 'bye')

(1, 2.8, 'hello world')

((2, 3, 4), (1, 2, 'hi'))

('hi', 'hello', 'bye')

hi

bye

8.9

2

v

22

(1, [9, 8, 7], 'world')

(1, [9, 8, 99], 'world')

(11, 22, 33, 44, 55, 66, 77, 88, 99)

(33, 44, 55)

(11, 22, 33, 44)

(55, 66, 77, 88, 99)

(55, 66, 77, 88)

(11, 22, 33, 44, 55, 66, 77, 88, 99)

(11, 22, 33, 44, 55, 66, 77, 88, 99)

True

False

False

True

apple

orange

grapes

banana

max value element : zara

max value element : 700

jack

26

1. Empty Dictionary: {}

2. Integer Keys: {1: 'apple', 2: 'ball'}

3. Mixed Keys: {'name': 'John', 1: [2, 4, 3]}

4. Using dict(): {1: 'apple', 2: 'ball'}

5. From Sequence: {1: 'apple', 2: 'ball'}

{1: 'apple', 2: 'ball'}

{'name': 'jack', 'age': 27}

{'name': 'jack', 'age': 27, 'address': 'downtown'}

{'name': 'john', 'code': 6734, 'dept': 'sales'}

dict_keys(['name', 'code', 'dept'])

dict_values(['john', 6734, 'sales'])

16

{1: 1, 2: 4, 3: 9, 5: 25}

{1: 1, 2: 4, 3: 9, 5: 25}

True

True

False

{1, 2, 3}

{1.0, (1, 2, 3), 'Hello'}

{1, 2, 3, 4}

{1, 3}

{1, 2, 3, 4}

The union of two sets is: {1, 2, 3, 4, 5, 6, 7, 8}

The intersection of two sets is: {4, 5}

nums1 - nums2: {1, 2, 3}

nums2 - nums1: {8, 6, 7}

{1, 2, 3, 4, 5, 6}

{3, 4}

{1, 2}